

Milestone-1 Report: NLP Foundation & Semantic Similarity

Objective: The objective of this milestone was to perform exploratory data analysis and basic NLP preprocessing on the competition dataset. The tasks involved text normalization, stop word filtering, Tf-Idf vectorization, cosine similarity computation, and MAP@3 metric evaluation.

Brief Dataset Description: The training dataset consists of multiple-choice questions containing a prompt, five answer options (A-E), and the correct answer label. The test data consist of same columns except the correct answer label. There are 2000 rows in the training dataset and 500 rows in the test dataset. No missing values were found in training and test dataset.

Text Preprocessing:

- Conversion of text to lowercase
- Removal of punctuation marks
- Tokenization using whitespace splitting
- Stop words removal using sklearn library

Tasks Performed:

- Text cleaning and preprocessing
- Tf-Idf Vectorization
- Cosine similarity computation
- map@3 metric evaluation

Answers to the milestone-1 questions:

Q1: 814

Q2: 859

Q3: 13

Q4: 2762

Q5: 0.2720

Q6: 13.55%

Q7: 1

Q8: 0.5

Q9: 0.42125

Q10: 0.29616

Conclusion: In this milestone, basic NLP preprocessing techniques such as text cleaning, stop word removal, Tf-Idf vectorization, and cosine similarity were applied to the dataset. The obtained results provide a baseline understanding of the data.